



DECODING CUSTOMER VALUE

A SQL-Driven Retention Strategy

Team Members

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Executive Summary

Building Brand Loyalty Without Sacrificing Growth



KEY TAKEAWAY



Customer loyalty is primarily driven by engagement and customer health, not discounts.
The highest-return strategy is to convert Future Champions into Premium Customers while gradually reducing promotional dependence

BUSINESS CHALLENGE



- Determine whether retention is driven by loyalty or discounts
- Identify the highest-value customer segments
- Develop a scalable retention strategy that improves profitability

MAJOR INSIGHTS

- Customer Health predicts retention better than promotional activity
- Future Champions are the strongest growth opportunity
- High-value states exist beyond promotion-heavy markets



STRATEGIC RECOMMENDATIONS

Invest

Prioritize Future Champions through loyalty programs and targeted engagement



Optimize

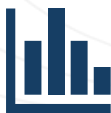
Gradually reduce discount intensity for Premium Customers



Expand

Focus growth investment on Tier-1 opportunity markets

EXPECTED BUSINESS IMPACT



Higher CLV



Reduced Discount Dependency



Improved Retention



Higher Marketing ROI

Building the Customer Intelligence Framework

A multidimensional framework was developed to measure customer value, loyalty, retention potential, and promotional dependency

FRAMEWORK DESIGN



DM-7: Full Price Loyalty Score (FPLS)

Formula:

$$0.5(SA_C) + 0.5(SD)$$

Purpose:

Measures loyalty without discounts.

RETENTION & HEALTH METRICS

DM-8: Customer Health Score (CHS)

Formula:

$$0.3(Active_norm) + 0.4(Eng_norm) + 0.3(RP_norm)$$

Purpose:

Overall relationship health.

Formula:

$$Rating_norm$$

Purpose:

Pure satisfaction.

DM-10: Satisfaction Risk Index (SRI)

Formula:

$$1 - Rating_norm$$

Purpose:

Disaffection risk.

DM-11: Retention Potential Score (RPS)

Formula:

$$0.5(CHS)$$

+

WHY THIS FRAMEWORK WORKS



KEY TAKEAWAY

Rather than relying on revenue alone, customers were evaluated through a multidimensional intelligence framework that links behavior, loyalty, value, and promotional dependence to business decisions

Customer Segmentation Reveals Unequal Growth Potential

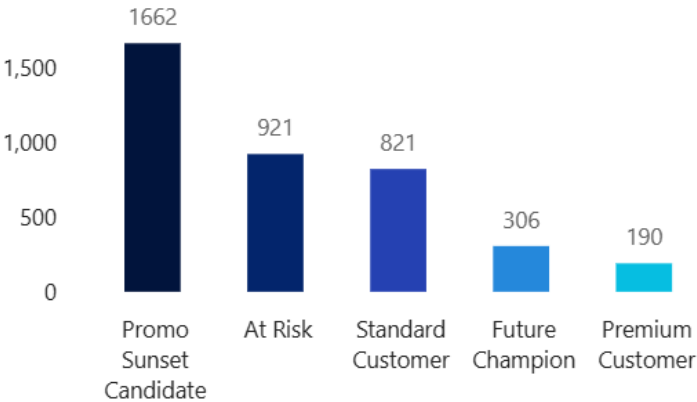
Not all customers contribute equally to long-term value, loyalty, and retention



CUSTOMER SEGMENTATION

STRATEGIC SEGMENT PLAYBOOK

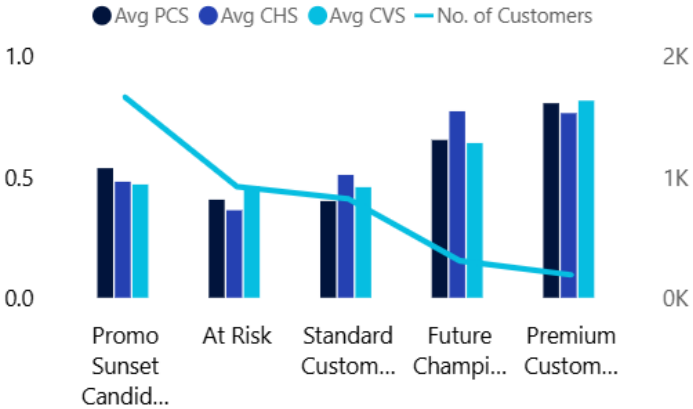
Customer Base Composition



Why Future Champions Matter

Future Champions combine near-premium customer quality with significantly greater growth headroom, making them the highest-return retention investment

Segment Quality Comparison



Segment Insights

- Premium Customers exhibit the strongest customer value and loyalty
- Future Champions demonstrate comparable quality with larger population scale
- Promo Sunset Candidates represent margin improvement opportunities through discount reduction
- At Risk customers require targeted recovery interventions rather than broad promotional spending.

Segment	Profile	Action
Premium Customers	Highest value & loyalty	Protect
Future Champions	High quality, scalable	Invest
Standard Customers	Moderate value	Develop
At Risk	Weak retention signals	Recover
Promo Sunset Candidates	Loyal but promo-independent	Reduce Discounts

KEY TAKEAWAY

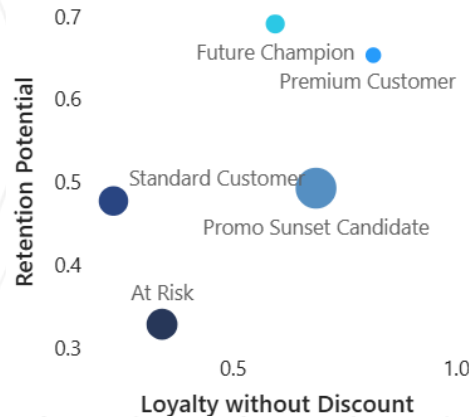
Customer value is concentrated within a small number of high-quality segments. Future Champions offer the strongest combination of customer quality, scalability, and retention potential, making them the primary growth investment segment

Loyalty Is Earned Through Engagement, Not Discounts

Retention is driven more by customer health and behavioral loyalty than promotional activity

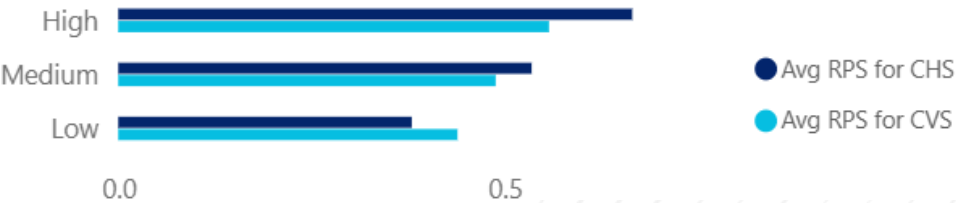


RETENTION vs LOYALTY MATRIX



High-value segments cluster in the high-retention, high-loyalty zone, indicating that engagement (not discounts) is the primary driver of customer persistence

RETENTION PREDICTIVE POWER



Customer Health demonstrates a stronger relationship with retention than promotional behavior, making it the most actionable retention indicator

KEY FINDINGS

Future Champions occupy the strongest retention-growth position



Customer Health Drives Retention



Loyalty Creates Persistence



Promotions Are Not the Primary Driver

BUSINESS IMPLICATIONS

Retain Future Champions, Protect Premium Customers, Optimize Promo Sunset Candidates

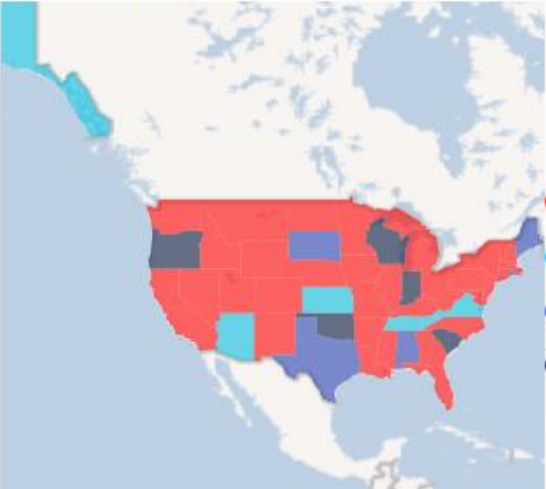
Growth Should Be Driven by Loyalty Instead of Discounts

Growth should be driven by high-value customers, high-opportunity markets, and reduced promotional dependence



GROWTH OPPORTUNITIES

Market Prioritization



- **Example Tier-1 Markets:** Alaska, Kansas
- **Example Tier-2 Markets:** Texas, Maryland
- **Focus:**
 - High Value
 - High Loyalty
 - Low Discount Dependency

IDEAL CUSTOMER PROFILE (ICP)

Ideal Customer Characteristics

- High Purchase Frequency
- Strong Customer Health
- High Loyalty Strength
- Moderate-to-High Purchase History
- Low Promotion Dependence
- Future Champion / Premium Characteristics



ACTION PLAN

EXPECTED IMPACT

Priority	Action	Outcome
Invest	Future Champions	Create more Premium Customers
Optimize	Reduce discounts for Premium Customers	Improve margins
Expand	Focus on Tier-1 states	Efficient growth

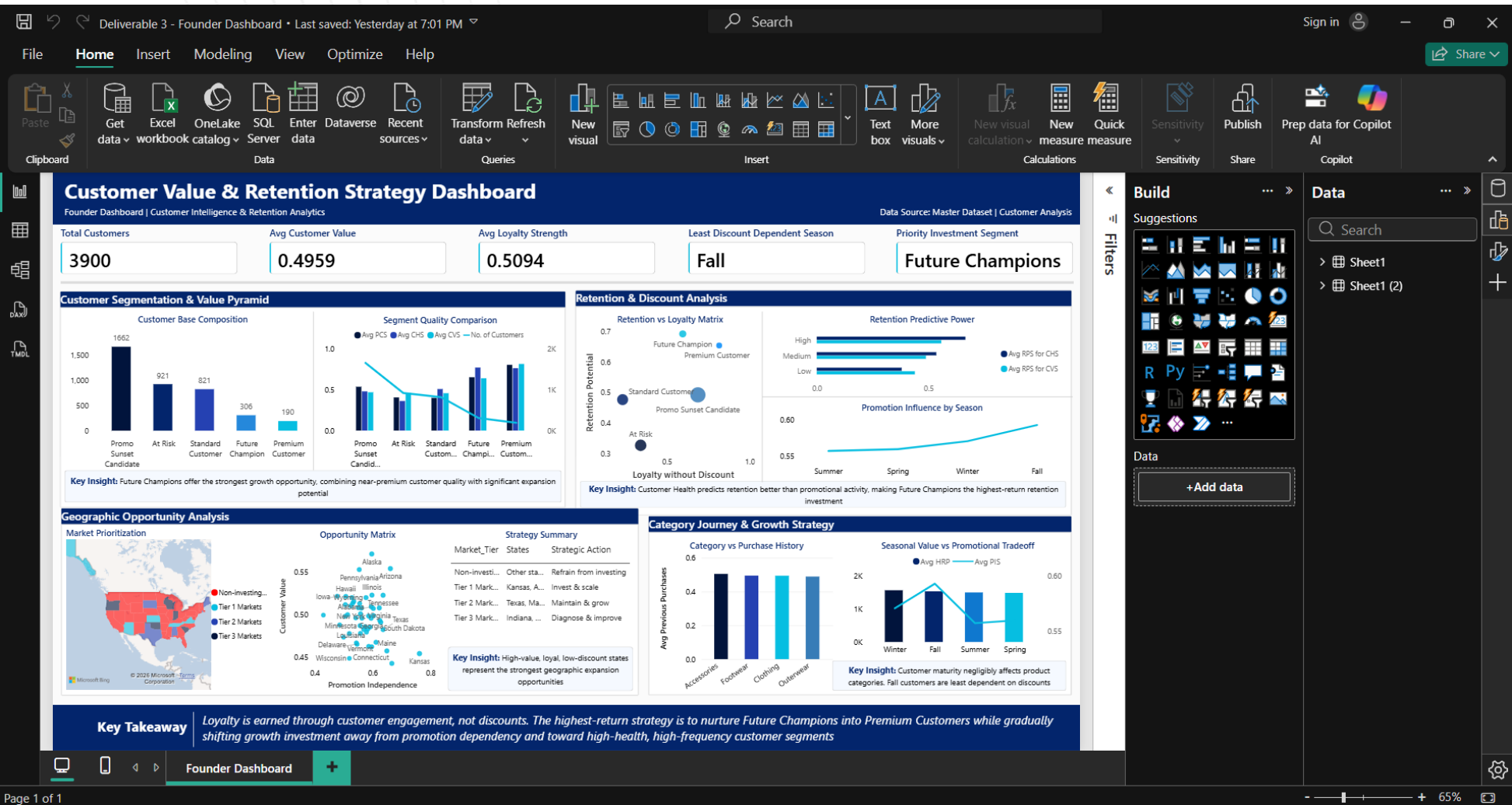
- **Short-Term**
 - Better targeting
 - Reduced promotional waste
- **Medium-Term**
 - Higher retention
 - Higher customer value
- **Long-Term**
 - Stronger profitability
 - Lower discount dependency

The most effective growth strategy is not increasing discounts, but systematically converting Future Champions into Premium Customers while expanding in high-affinity markets



THANK YOU

INTERACTIVE DECISION DASHBOARD



Dashboard Purpose

- Monitor customer value, retention, loyalty, geography, and growth opportunities
- Enable segment-level strategic decisions
- Translate analytics into business actions



SQL-BASED CUSTOMER INTELLIGENCE ENGINE

H-2: Customers with higher purchase frequency generate higher customer value

- Metrics used:
CVS, PCS, Freq_norm, Frequency of Purchases
- Insight Path
Frequency
↓
CVS / PCS
↓
Does engagement drive value?

Query:

```
SELECT
  `Frequency of Purchases`,
  ROUND(AVG(CVS),4) AS Avg_CVS,
  ROUND(AVG(PCS),4) AS Avg_PCS,
  COUNT(*) AS Customers
FROM `master dataset`
GROUP BY `Frequency of Purchases`
ORDER BY Avg_CVS DESC;
```

Result:

Frequency of Purchases	Avg_CVS	Avg_PCS	Customers
Weekly	0.5952	0.6221	539
Fortnightly	0.5594	0.5772	542
Bi-Weekly	0.5333	0.543	547
Monthly	0.4945	0.5011	553
Quarterly	0.4749	0.4701	563
Every 3 Months	0.4305	0.4194	584
Annually	0.3953	0.3815	572

Metric	Purpose
CVS	Customer Value
CHS	Customer Health
FPLS	Loyalty Strength
RPS	Retention Potential
PCS	Premium Potential
PIS	Promotion Dependence

Outcome

Converted raw transactional data into decision-ready customer intelligence metrics

CUSTOMER SEGMENTATION METHODOLOGY

• Query:

```
SELECT
`Customer ID`,
ROUND(CVS,4) AS CVS,
ROUND(PCS,4) AS PCS,
ROUND(HRP,2) AS HRP
FROM `master dataset`
ORDER BY CVS DESC
LIMIT 20;
```

• Result:

Customer ID	CVS	PCS	HRP
1512	0.9315	0.7473	4368
4	0.9314	0.7772	4410
1286	0.9298	0.7133	4455
2709	0.9258	0.8774	4136
1738	0.9208	0.8646	4900
1329	0.92	0.7243	3880
898	0.9193	0.7614	4186
1889	0.919	0.8862	4272
2803	0.9177	0.8403	4200
2663	0.9174	0.8379	4312
1436	0.9136	0.7308	3895

Customer_Segment	Customers	Avg_CVS	Customer_Percentage
Premium Customer	189	0.8175	4.85
Future Champion	306	0.6408	7.85
Promo Sunset Candidate	1693	0.4695	43.41
At Risk	892	0.463	22.87
Standard Customer	820	0.458	21.03

• Evidence:

Customer Distribution		
Segment	Customers	% of Customers
Premium Customer	189	4.85%
Future Champion	306	7.85%
Standard Customer	820	21.03%
At Risk	892	22.87%
Promo Sunset Candidate	1693	43.41%
Total Value Contribution		
Segment	Total CVS	
Promo Sunset Candidate	794.89	
At Risk	413.04	
Standard Customer	375.55	
Future Champion	196.1	
Premium Customer	154.5	

Key calculations:

- Premium Customers + Future Champions = **12.70% of customers**
- Their combined Total CVS = **350.60**
- Total CVS across all customers = **1934.08**
- Contribution = **18.13% of total customer value**

Additionally:

- The largest segment (Promo Sunset Candidate) represents **43.41% of customers** and contributes **41.10% of total value**.
- Value contribution broadly follows customer distribution rather than being concentrated among a small elite group.

Segment

Logic

Premium Customer	High CVS + High Loyalty
Future Champion	High PCS + Strong Health
Standard Customer	Moderate Scores
At Risk	Weak Retention Signals
Promo Sunset Candidate	Loyal but Discount Independent

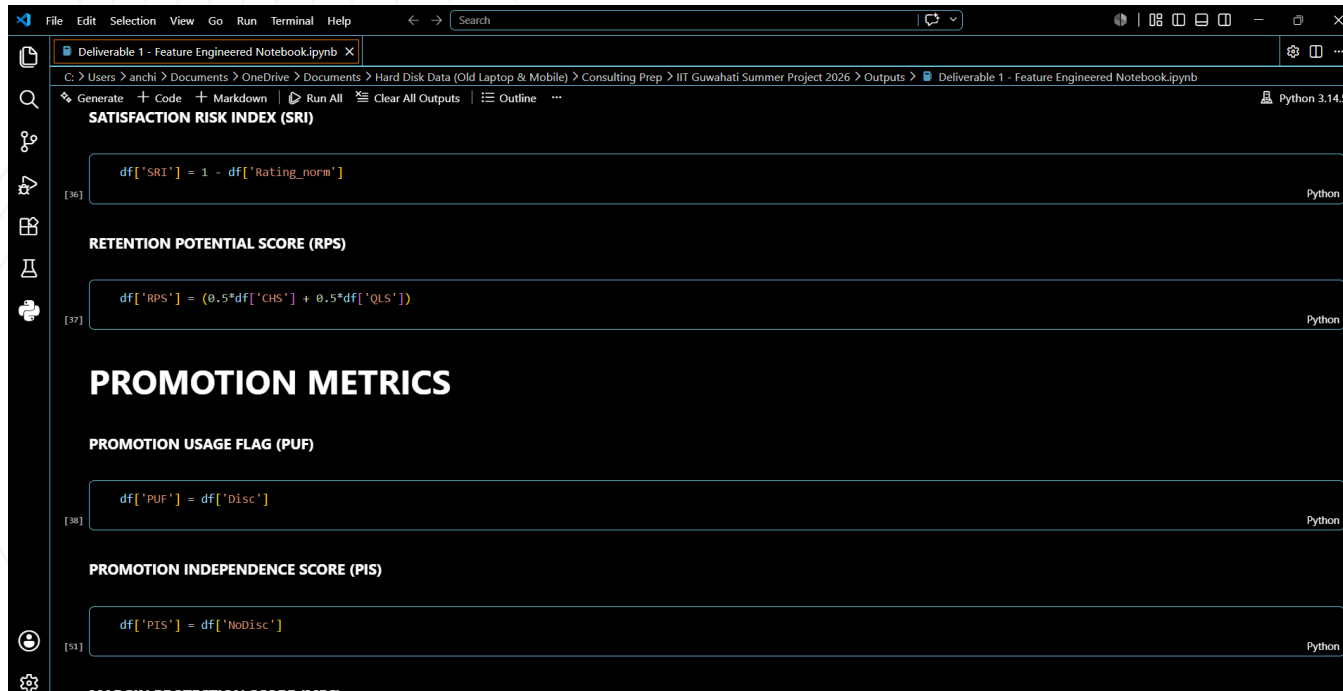


HYPOTHESIS TESTING SUMMARY

Hypothesis	Result
Loyalty drives retention more than discounts	Accepted
Future Champions are strongest growth segment	Accepted
Product categories determine loyalty	Rejected
Product categories determine maturity	Rejected
Geographic value differences exist	Partially Accepted
Customer Health predicts retention better than promotions	

Strategic recommendations were based only on validated hypotheses

STATISTICAL & EXPLORATORY ANALYTICS



```
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Generate + Code + Markdown | Run All Clear All Outputs | Outline ... Python 3.14.5

SATISFACTION RISK INDEX (SRI)
df['SRI'] = 1 - df['Rating_norm'] [36] Python

RETENTION POTENTIAL SCORE (RPS)
df['RPS'] = (0.5*df['CHS'] + 0.5*df['QLS']) [37] Python

PROMOTION METRICS
PROMOTION USAGE FLAG (PUF)
df['PUF'] = df['Disc'] [38] Python

PROMOTION INDEPENDENCE SCORE (PIS)
df['PIS'] = df['NoDisc'] [51] Python
```

Python Contributions

- *Feature engineering*
- *Correlation analysis*
- *Metric validation*
- *Exploratory pattern detection*